GOOGLE CLOUD AI BUILDER CUP 2026 · THEME: BFSI

Rakshak

Stop the scam at the moment of payment.

A guardian agent inside the bank app. It hears the coercion on the call, checks who is being paid, and pauses the transfer with a voice in the customer's own language.

Team Rakshak · [add member names, roles, contact]

https://rakshak-223197895518.asia-southeast1.run.app · github.com/devaramanelohith/rakshak-agent

THE PROBLEM

The money leaves while the scammer is still talking

81.8%

of Singapore's 2025 scam cases were self-effected transfers: the victim paid.

Singapore Police Force, Feb 2026

₹22,495 cr

lost to cyber fraud in India in 2025. About 24% was recovered.

Moneylife / MHA data

123,672

digital-arrest cases in 2024, ₹1,935.51 crore lost.

Parliamentary data

Under MAS's Shared Responsibility Framework the liability for missed real-time controls sits with the bank. Nothing today combines what is said on the call with who is being paid.

police.gov.sg · moneylife.in · MHA parliamentary replies

WHY TODAY'S DEFENCES MISS IT

Everyone sees one piece. Nobody joins them when the money moves.

Defence	Sees	Acts	Misses
Pixel scam detection (on device)	Speech patterns	Warn only	No payee knowledge, no hold, nothing reaches the bank
Call-state signals (FICO Scam Signal, BioCatch)	A call is in progress	Risk score at payment	Cannot tell a daughter asking for fees from a fake CBI inspector
Bank fraud engines	Payer, payee, amount	Block or step-up	The coerced victim passes every step-up: they want to pay
Post-debit SMS, 1930 helpline	The completed debit	Notify, file	Mule accounts empty within hours; ~24% recovery
Rakshak	**Call semantics x payee mule risk x profile**	**Graded hold at authorisation**	**Voice in the customer's language, trusted-contact alert, case + 1930 referral, one-tap override**

Sources in docs/DEMO_STRATEGY.md and the market check

WHAT RAKSHAK DOES

Four moments, one guardian

With the customer's opt-in, Rakshak scores the caller's coercion live and, when Pay is tapped, fuses it with the payee's mule risk to pause the transfer.

1 · LISTEN

Per-speaker transcription

Live transcript, 8 coercion signals, risk meter and reasons in Hindi, Tamil, Kannada or English.

2 · WHISPER

Private warning

At risk 0.85 the customer hears "Rakshak: careful". The caller never does.

3 · HOLD + VOICE

Pay is tapped

Decision in under 100 ms. 30-minute hold, voice barge-in over the caller, payee evidence, typed-amount override.

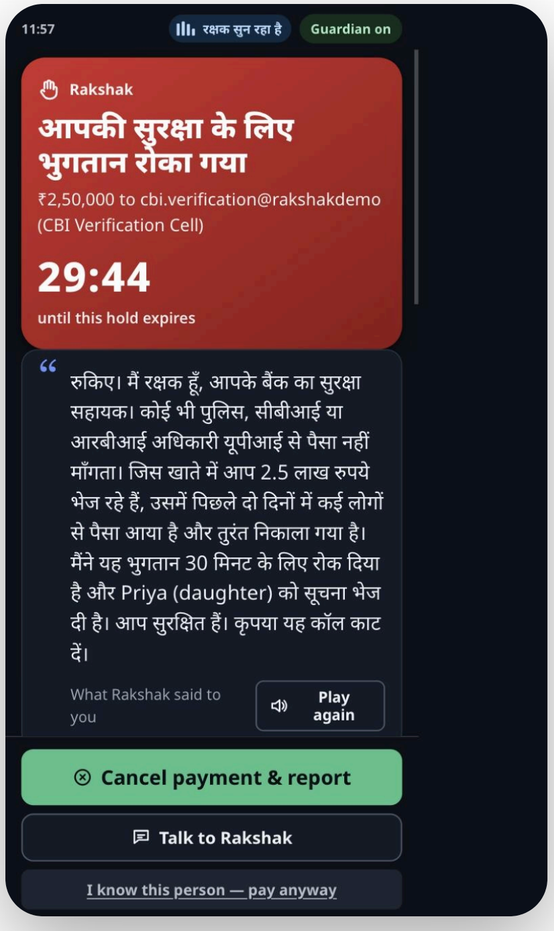
4 · ALERT

Trusted contact + desk

Daughter's phone chimes. Fraud desk gets the case, agent trace and a drafted 1930 referral.

THE PAYMENT MOMENT

What the customer sees one second after Pay



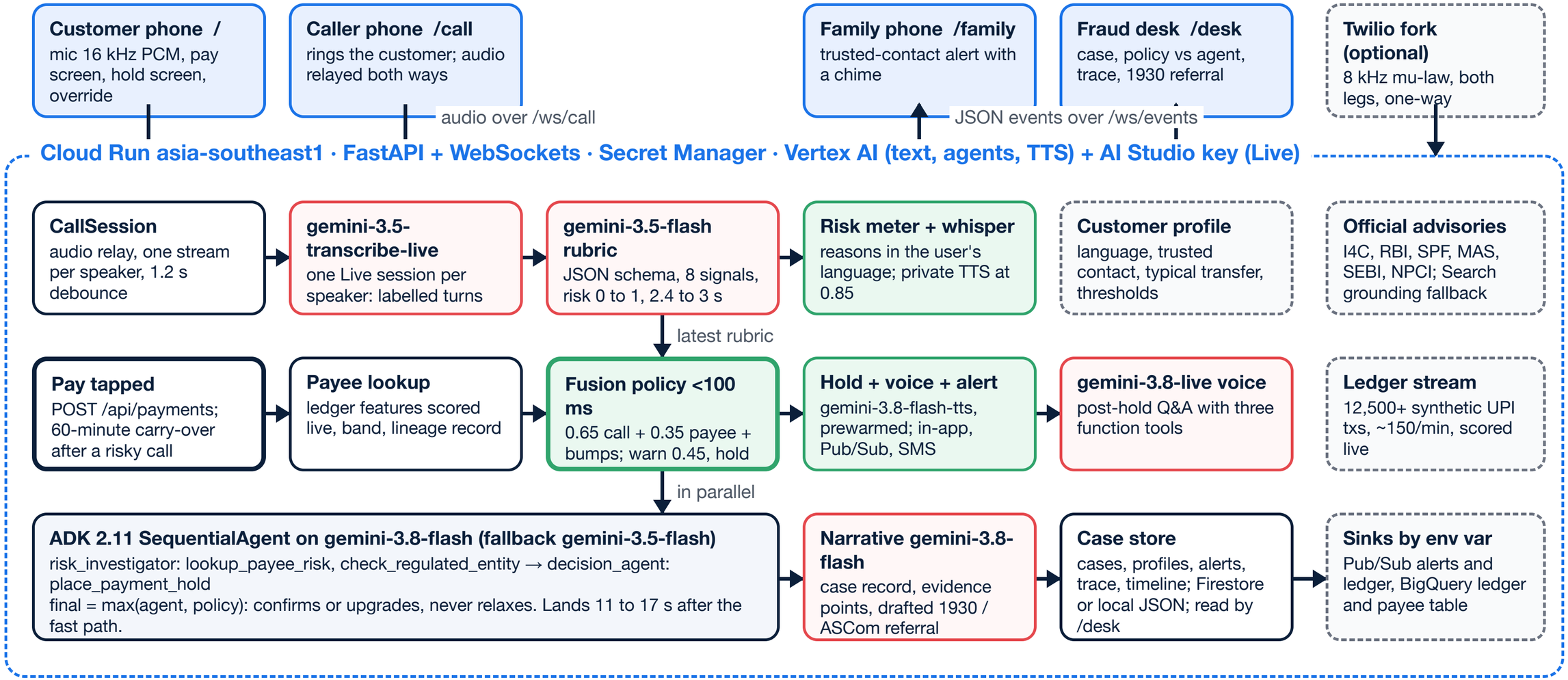
Capture from https://rakshak-223197895518.asia-southeast1.run.app/ after tapping Pay on the "Parcel from Taiwan" script. 30:00 countdown, Hindi spoken text, payee card, "Priya (daughter) has been notified", three exits.

Measured on the public UPI hold event 20 ms to 1.2 s after Pay; voice in 100 ms when prewarmed

- HOLD "Paused for 30 minutes", spoken in Hindi over the ducked caller audio

ARCHITECTURE

Decide fast, explain slower: two paths, one floor



Devices · Gemini models · Actions · Agents · Data

Tools and the payee lookup read the advisory table, the ledger features and the profile.

GOOGLE CLOUD AI USAGE

Five Gemini models, ADK, Cloud Run: each with a job

Model or service	Where	Why
gemini-3.5-transcribe-live	Live API, one session per speaker	Labelled caller/user turns without diarization
gemini-3.5-flash	Coercion rubric, strict JSON schema	2.4 to 3 s with minimal thinking; clean Devanagari, Kannada, Tamil output
gemini-3.8-flash	ADK investigator and decision agents, narrative, Search grounding	Best tool use; latency is off the critical path
gemini-3.8-flash-tts	Voice barge-in and whisper, prewarmed per amount	Natural Hindi prosody; 100 ms when cached
gemini-3.8-live	Post-hold voice conversation with function calling	Customer asks "why?" and hears the answer
Google ADK 2.11	SequentialAgent, Runner, function tools	Traceable multi-agent decision with a policy floor
Cloud Run + Secret Manager + Vertex AI	asia-southeast1, WebSockets, session affinity	One service, no key in the image
Firestore, BigQuery, Pub/Sub	Cases, ledger sink and payee SQL, alert fan-out	Switched on by env var; same code path

/api/health reports the live model ids

THE AGENT

An ADK pipeline that can only make the bank stricter

1 risk_investigator

- `lookup_payee_risk`: ledger features, band, lineage
- `check_regulated_entity`: official-advisory table (I4C, RBI, SPF, MAS, SEBI, NPCI), then Search grounding
- Writes a 120-word findings note

2 decision_agent

- `place_payment_hold`: the human-in-the-loop action node
- Message written in the customer's language with concrete evidence

Fast path first. The deterministic policy acts in under 100 ms. The agent lands 11 to 17 s later with evidence and a trace.

Policy floor. final = max(agent, policy). The agent confirms or upgrades, never relaxes a hold. A failed agent leaves the policy in force.

Human in the loop. A hold is a pause: one-tap override with a typed amount, voice Q&A, 30-minute expiry. Every step is on the case timeline.

app/agents/pipeline.py · app/agents/tools.py · app/agents/advisories.py

WHERE THE DATA COMES FROM

Payee risk is the bank's own ledger, with lineage

Feature	Source in production	In this prototype
fan_in_48h, cash_out_ratio_7d, dormant_days, account_age	Bank or PSP ledger, scored by sql/payee_risk.sql on a schedule	Synthetic UPI ledger stream: 12,500+ transactions over 7 days, ~150 new per minute, 64 payees, same formula live
fri_level	DoT Financial Risk Indicator feed	Static registry value
linked_complaints_1930	I4C suspect registry via the bank	Static registry value
kyc_tier	Bank KYC	Static registry value

Score = 0.30 fan-in + 0.20 cash-out + 0.15 dormancy + 0.15 FRI + 0.15 complaints + 0.05 new account. Many payers alone is a shop; the combination is a mule.

Every payee card carries a lineage record: source, transactions in window, computed-at time, which fields are registry values, which sinks (Pub/Sub, BigQuery) mirror the stream.

Synthetic ledger in this prototype; no bank data was used. Production: Pub/Sub to BigQuery, scheduled SQL, Firestore lookup.

EVALUATION

50 labelled calls, including the ones written to fool it

	Pred scam	Pred benign
Scam (25)	25	0
Benign (25)	0	25

Call-level detection at the 0.70 hold threshold, gemini-3.5-flash rubric

- Hindi, Hinglish, English, Singapore English, Kannada, Tamil
- 10 hard negatives: real police, hospital deposit, harsh landlord, SEBI adviser
- 10 rubric-gaming scams: no "arrest", "OTP" or "safe account" said aloud
- Payment-decision accuracy 0.974: the one miss is a by-design **warn** on an ambiguous job-scam call
- System tests: smoke_ws (API + Live voice), smoke_live_audio (real Hindi speech through the relay, risk 0.9 to 1.0, whisper fired), two-way relay

1.00
precision

1.00
recall / F1

0.974
payment-decision accuracy

0.932
per-signal accuracy (8 x 50)

2.4 s
mean rubric latency, uncontended (5.3 s contended)

scripts/eval.py · data/eval/calls.json · docs/EVAL_RESULTS.md

PRIVACY BY DESIGN

The customer chooses the tier; it works at every tier

Tier	Rakshak uses	Leaves the phone	Status
0: no audio	Call in progress + pay screen + payee risk + profile	Only the payment request	Design; policy runs with call risk 0
1: on-device transcript	Rolling 60 s transcript on the device; scores sent only when Pay is attempted	Scores + last window, at Pay only	Design; needs an OS or OEM audio channel
2: cloud, lawful channel	Live transcription for in-app calls, telco forks or OS-vendor feeds	Audio stream (not retained), transcript on the case	This prototype

- Opt-in with a visible "Rakshak ON" pill
- Raw audio never written to disk
- The caller never hears the whisper
- Override always available; holds expire

The cellular-call answer. No third-party app can tap a cellular call on Android 10+ or iOS. So Rakshak runs bank-side: in-app calls, telco audio forks (FICO Scam Signal precedent) or OS-vendor feeds. Google is the one company that can wire Pixel scam detection to a GPay hold; Rakshak is the bank-side half.

IMPACT AND BUSINESS CASE

Who pays, what it costs, how it scales

Who buys

Banks, UPI and PayNow apps that carry the liability under MAS's Shared Responsibility Framework and RBI's customer-protection rules, and the recovery cost of every missed scam.

What they get

Fewer authorised-push-payment losses, an auditable case with a drafted 1930 or ASCom referral per intervention, and a customer who was protected instead of blamed.

Cost per call

Two Live sessions, 10 to 20 flash rubric calls, one cached clip, one ADK run: cents per guarded call at flash-tier prices, versus a ₹2 lakh transfer.

Scale path on Google Cloud

- Sessions and cases in Firestore (multi-instance, durable)
- Alerts fanned out through Pub/Sub to SMS, push, dialler
- Ledger to BigQuery; sql/payee_risk.sql as a scheduled query; materialised for sub-10 ms lookup
- ADK pipeline on Vertex AI Agent Runtime; profiles in Memory Bank
- Cloud Run: WebSockets, session affinity, min instances; stateless beyond the live call

> Ships as an SDK in the bank app (consent, mic, pay hook) plus the bank-side service; a telco variant streams the call like the Twilio bridge.

Pricing model stated as a hypothesis: per protected account per month, or per intervention

ROADMAP AND LIMITATIONS

What is real, what is synthetic, what comes next

Honest limitations

- Ledger and calls are synthetic; no bank partner yet
- Twilio bridge is one-way: the voice plays on the app, not into the PSTN call
- Kannada and Tamil voice not reviewed by native speakers
- Tier 1 on-device is a design, not code
- Prototype endpoints are open; single always-on instance for the demo
- The agent's own decision lands 11 to 17 s after the fast path

Next 90 days

- Real mule features from a bank ledger through sql/payee_risk.sql
- ADK pipeline on Vertex AI Agent Runtime; profiles on Memory Bank
- Bidirectional telco stream so the barge-in reaches the call
- Native-speaker review of Kannada and Tamil
- Bank pilot behind Firebase Auth and App Check
- Human-labelled real call corpus for the evaluation

> **The ask:** one bank fraud team and one telco to pilot.

TRY IT AS A JUDGE IN 60 SECONDS

Rakshak: stop the scam at the moment of payment

1. Open `rakshak-223197895518.asia-southeast1.run.app/?session=judge&user=demo-user`, finish the three onboarding taps
2. Tap **Test an incoming call**, choose **Parcel from Taiwan**, tap **Auto-play script**; watch the gauge and the Hindi reasons
3. When the toast says the caller demands payment: **Open UPI pay**, then **Pay**
4. See the hold, hear Rakshak in Hindi, then open `/desk` for the agent trace and the 1930 referral

Live: https://rakshak-223197895518.asia-southeast1.run.app

Code: https://github.com/devaramanelohith/rakshak-agent

Team Rakshak · [names and contact]